

Accounts Merge (LeetCode 721)

| Logical Breakdown (My Notes)

1. Approach: basically dsu lagaynge

2. Grouping Emails: sbse pehle har email ka account holder dekhenge using a unordered map.

- pehli baar dikha to account holder
- doosri baar dikha to union it with account holder index (*1 se kyunki 0 pe name he*).

3. Mapping Leaders: ab har mail ke paas ek unique leader index he.
multiple email ke paas ek hi leader ho skta isliye ab ek map bnayenge jisme sirf leader->all mails

4. Final Formatting: ab bas format me store krna baccha he. Sort the mapped emails and append to the result.

| Source Code

```
#include<bits/stdc++.h>
using namespace std;

class dsu{
public:

    vector<int> parent;
    vector<int> size;
    dsu(int n){
        parent.resize(n);
        size.resize(n,1);

        for(int i=0;i<n;i++){
            parent[i]=i;
        }
    }

    int findparent(int node){

        if(parent[node]==node){
            return node;
        }
    }
}
```

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        parent[node]=findparent(parent[node]);
        return parent[node];
    }

    void unionit(int u,int v){
        int pu=findparent(u);
        int pv=findparent(v);

        if(pu==pv) return;

        if(size[pu]<size[pv]) swap(pu,pv);

        parent[pv]=pu;
        size[pu]+=size[pv];
    }
};

class Solution {
public:
    vector<vector<string>> accountsMerge(vector<vector<string>>& accounts) {
        //basically dsu lagaynge
        //sbse pehle har email ka account holder dekhenge
        //using a unordered map
        //pehli baar dikha to account holder
        //doosri baar dikha to union it with account holder index

        int n=accounts.size();
        dsu ds(n);

        unordered_map<string,int> mp;

        for(int i=0;i<n;i++){
            for(int j=1;j<accounts[i].size();j++){ //1 se kyunki 0 pe name he

                string mail=accounts[i][j];

                if(mp.find(mail)==mp.end()){
                    mp[mail]=i;
                }

                else{
                    ds.unionit(mp[mail],i);
                }

            }
        }

        //ab har mail ke paas ek unique leader index he
        //multiple email ke paas ek hi leader ho skta
        //isliye ab ek map bnayenge jisme sirf leader->all mails

        unordered_map<int,vector<string>> finally;
        for(auto &it:mp){

            string mail=it.first;
            int idx=it.second;

            int boss=ds.findparent(idx);

```

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        finally[boss].push_back(mail);

    }

    //ab bas format me store krna baccha he
    vector<vector<string>> ans;
    for(auto &it:finally){

        int bossidx=it.first;
        vector<string> curr;

        curr.push_back(accounts[bossidx][0]);
        vector<string> mails=it.second;
        sort(mails.begin(),mails.end());
        for(string m:mails){
            curr.push_back(m);
        }

        ans.push_back(curr);
    }

    return ans;
}

};

```